



IRON ORE 2030

THE SUPPLY WAVE



CASE ANALYSIS: New Mines, Structural Surplus, and the Price Reckoning

Scope: Global Seaborne Iron Ore Supply | 2025-2030

April 2026 | Research & Analytics | Adarsh Sahu



What This Study Addresses: Five Key Questions

1. How big is the supply wave?

Who is adding tonnes, when, and at what cost. Three-wave pipeline 2025-2032.

2. Gross vs. net - the critical distinction

Rio's ~130 Mtpa Australian replacement programme adds ~zero net seaborne supply. Most analyses miss this.

3. What does Simandou actually add?

Rio's 5-10 Mt guidance = SimFer equity only. Combined WCS+SimFer = 25-35 Mt in CY2026. 3-4x more.

4. India FY2026 = 310 mnt - lagging slight against steel production growth

FY2025 = 289 Mt. FY2026 = 310 Mt. Different fiscal years. Matters for supply-demand modelling.

5. Where does price settle by 2030?

Base case \$75-82/dmt (IODEX 61% Fe). Grade bifurcation: DR-grade resilient; 58% Fe discount widens.



Global iron ore fines index shifted from 62% Fe to 61% Fe specification effective January 2, 2026. Transitional 61/62% basis spread ~\$1.5-2/t published.

Price Interpretation

All 2026+ IODEX readings = 61% Fe basis. Historical 2021-2025 was 62% Fe basis. The ~\$1.5-2/t difference is NOT material at a 5-year horizon. This study's 2030 scenario ranges (\$65-100/t) are unchanged.

Grade Spread Re-referencing

65%+ Fe premium: marginally wider in absolute \$ vs. old 62% basis (lower base). 58% Fe discount: marginally narrower. Both directional trends unchanged. Simandou (65.3% Fe) is now 4.3% above the new 61% benchmark.

Why the Benchmark Changed

Pilbara ore grades have been declining for a decade. By 2025, the 62% Fe spec no longer reflected what was predominantly traded - PBF shipments tracked closer to 61% Fe. The shift reflects market reality.

BHP-CMRG + Benchmark Coincidence

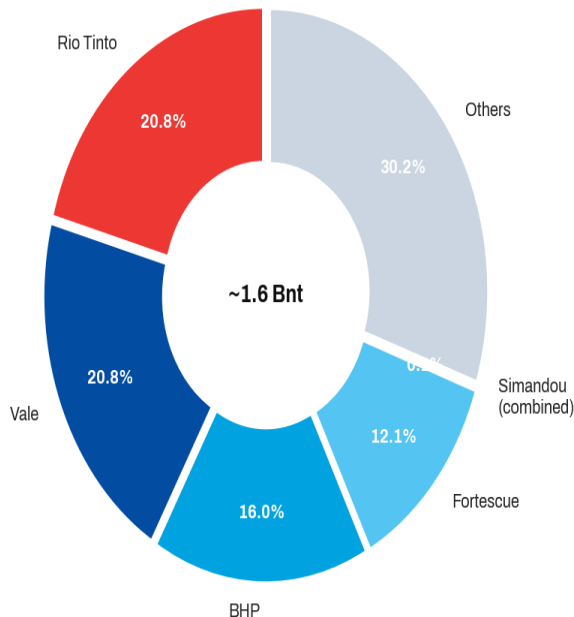
The benchmark shift (Jan 2, 2026) and BHP-CMRG dispute coincided. Jimblebar fines (62-63% Fe) traded at -10-11% discount to the new 61% Fe index during the ban vs. normal -7-8%. Quality-adjusted pricing became more complex.

Seaborne Market Structure: Simandou Enters as 5th Major Exporter by 2030

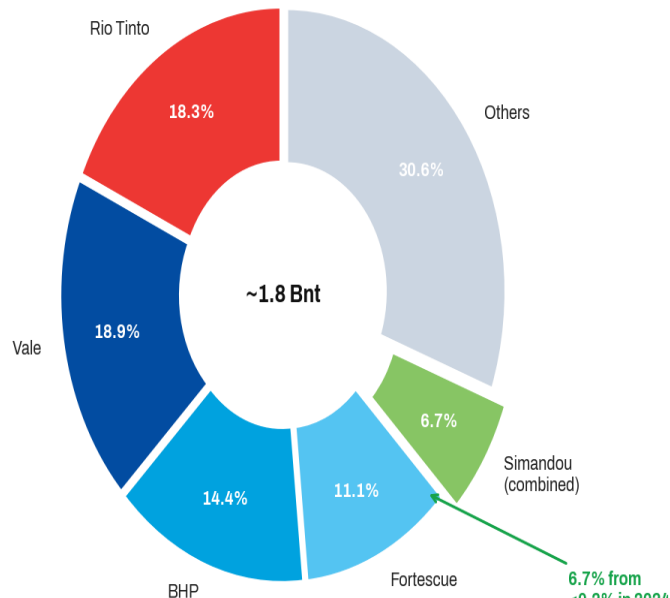
Seaborne Iron Ore Market Share: 2024 vs. 2030 Base Case

Simandou (Guinea) rises from <0.2% to ~6.7% by 2030. Big 4 concentration dilutes.

2024 Estimate



2030 Base Case



~1.58 Bnt/yr

Seaborne trade 2024 base

~6.7%

Simandou 2030E share
(from <0.2%)

177.5 Mt

China port inventory Apr 2 2026 –
near-record price buffer

-10 Mt

China seaborne imports drop
expected
in 2026 (~1.25 Bnt)

Source: BigMint analysis; operator production data. 2024 base ~1,580 Mt; 2030E base ~1,800 Mt.

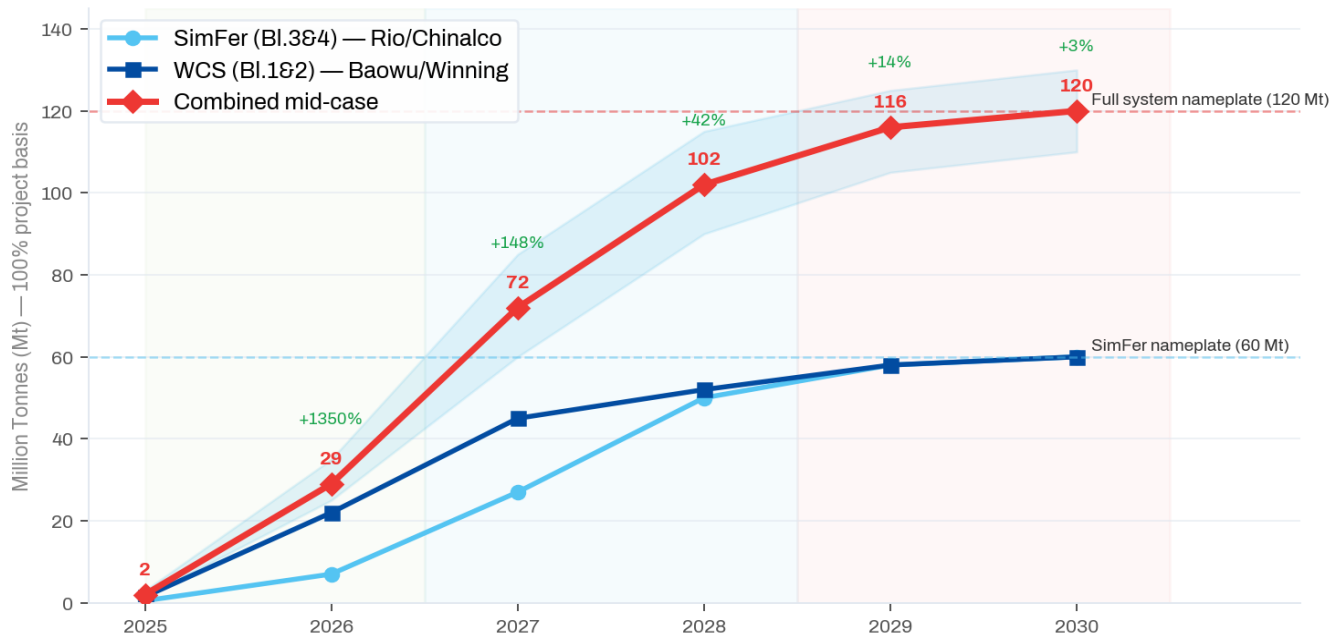
Top 10 Producer League: Verified from Primary Filings (mnt)

#	Producer	Country	CY2025 Output	2026 Guidance	C1 (\$/t)	Tier	Key Verified Data Point
1	Rio Tinto	Aus/Guinea	336.6	343-366	~\$24/wmt	T1	Q1 2026: 78.8 Mt Pilbara (+13% YoY). SimFer Q1=0.6 Mt (100% basis). Cash cost guidance \$23.5-25/wmt.
2	Vale	Brazil	336.1	335-345	\$21.3/t	T1	C1 \$21.3/t explicitly stated in FY2025 6-K (SEC, Jan 28 2026). 7-yr high.
3	BHP (FY basis)	Australia	~260	258-269	~\$18/t	T1	290 Mt wet 100% basis (record). FY2026 guidance equity basis. BHP-CMRG ban lifting Apr'26.
4	Fortescue (FY)	Australia	198.4	195-205	\$19.04/wmt	T1	RECORD FY2025 +4% YoY. C1 \$19.04/wmt stated H1 FY2026. Iron Bridge 67% Fe ramping.
5	WCS/Baowu	Guinea	~1.5	~15-25	~\$24*	T1*	First shipper Dec 2 2025. Baowu 51% stake Jan 2026. WCS+SimFer combined = 25-35 Mt CY2026.
6	SimFer (Rio)	Guinea	~0.5	5-10	~\$25*	T1*	Q1 2026 = 0.6 Mt 100% basis (Rio Q1 6-K Apr 21 2026). Rio guidance SimFer only.
7	NMDC	India	53 (FY)	53-56	~\$32*	T2	Record prod. FY2026 +21% YoY (Apr 2026). Primarily captive domestic supply.
8	ArcelorMittal Lib.	Liberia	~9.5	~18	~\$35*	T2	Nimba expansion. 245% increase CY2024 →2026E. Net new seaborne supply.

* = Author proxy. FY = July-June. BHP/FMG on FY basis. Simandou T1* = structurally new seaborne supply.

Simandou Production Ramp 2025–2030 (Mt, 100% Project Basis)

SimFer Q1 2026 actual = 0.6 Mt (100% basis). CY2026 combined working est. 25–35 Mt (mid 29 Mt). % labels = YoY change.



Source: Rio Tinto Q1 2026 6-K (Apr 21 2026); SMM Iron & Steel; BigMint analysis. WCS estimated.

~3.3 Bnt

Reserves at ~65.3% Fe avg - highest-grade new seaborne entrant

25-35 mnt

Working CY2026 combined est.
WCS+SimFer (not Rio's 5-10 Mt equity)

0.6 mnt

SimFer Q1 2026 actual (100% basis) per
Rio Q1 6-K Apr 21 2026

670 km

TransGuinean railway - 12 stations, 200+ bridges, 4 tunnels

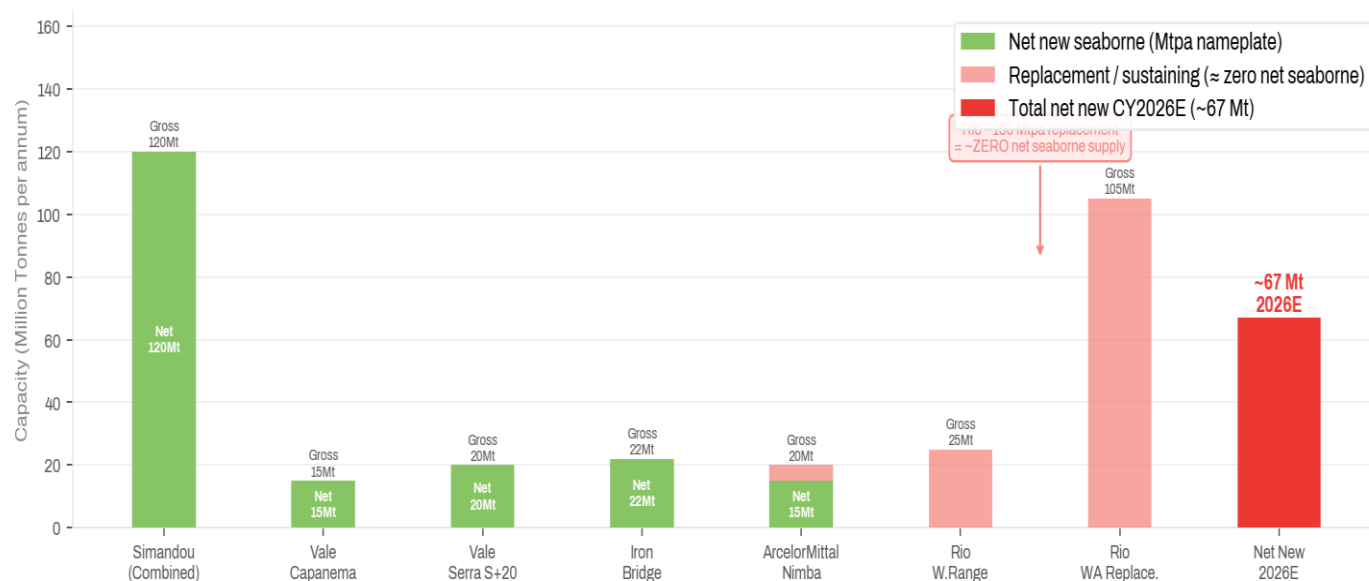
65.3% Fe

Product grade is 4.3% above new IODEX
61% Fe benchmark

CRITICAL: Rio Tinto's "5-10 Mt" = SimFer equity share ONLY (Blocks 3&4, ~53% stake). WCS (Blocks 1&2, Baowu 51%) first shipped Dec 2, 2025 and is ramping FASTER. Combined CY2026 = 25-35 Mt. Using Rio's figure alone understates 2026 impact 3-4x.

Gross vs. Net Seaborne Supply Additions: The Critical Distinction

Replacement mines offset depletion → net ≈ zero. Only genuine new seaborne tonnes count in surplus arithmetic.



Source: Operator filings (Rio Tinto, Vale, Fortescue investor presentations). BigMint analysis.

Why Net ≠ Gross

● Replacement mine

New mine offsets depleting pit at same operator. Net seaborne addition: ~ZERO.

● Rio ~130 Mtpa programme

Western Range, West Angelas, Brockman Syncline 1 - all replacement. Maintain ~327 Mt Pilbara.

● Genuine net additions CY2026

Simandou ~29 Mt + Vale ~15 Mt + Liberia ~12 Mt + Iron Bridge ~11 Mt = ~67 Mt.

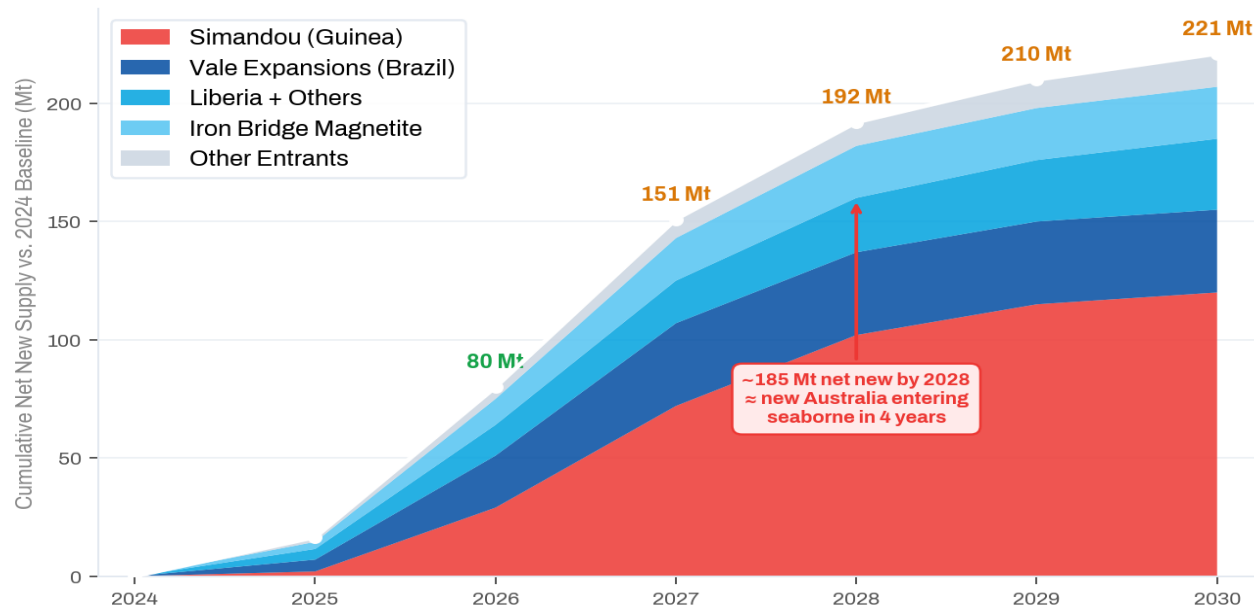
● The overstatement risk

Counting Rio's replacement tranche as new supply overstates the surplus by ~100+ Mt nameplate.

Cumulative Net New Seaborne Supply + The Inventory Buffer Mechanism

Cumulative Net New Seaborne Supply vs. 2024 Baseline (Mt) — Base Case

Replacement-adjusted. Simandou dominates. ~67 Mt net new in CY2026 alone.



2-Stage Price Transmission

STAGE 1 —> 2025-2027

Net new tonnes arrive (~67 Mt in CY2026). Port inventory at 177.5 Mt (Apr 2) absorbs arrivals. 6-12 month price discovery lag. CURRENTLY ACTIVE.

STAGE 2 —> 2027-2029

Inventory normalises toward 130-145 Mt. Price pressure becomes direct. Base case \$75-82/dmt by 2030. The price reckoning.

~67 Mt

Net new seaborne CY2026

~185 Mt

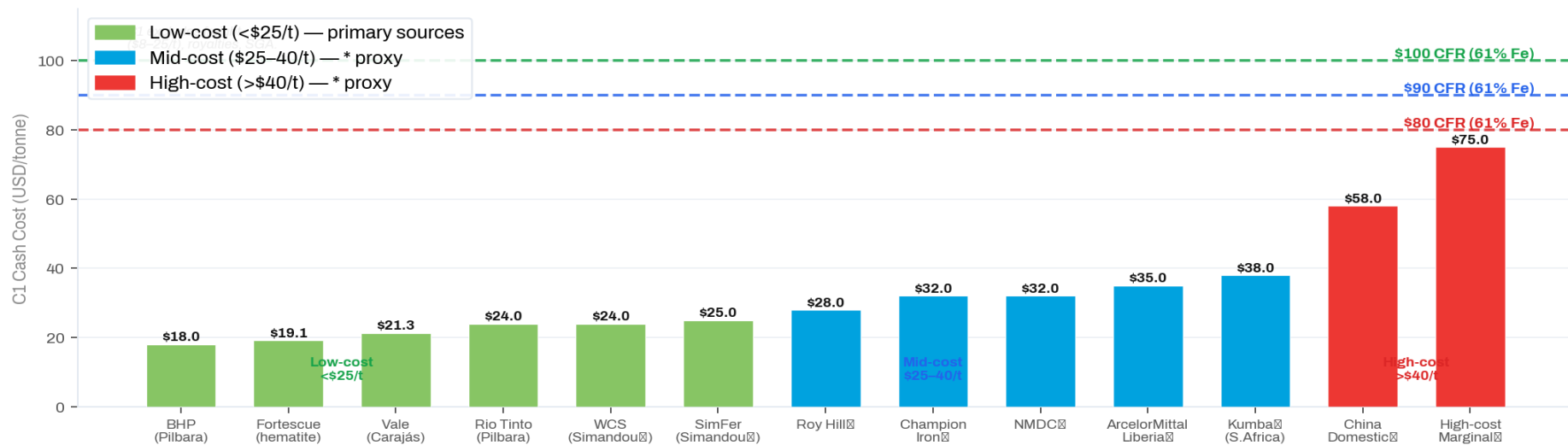
Cumulative by 2028

177.5 Mt

China port inventory Apr 2 2026

Proxy C1 Cash Cost Curve: Global Iron Ore Producers (USD/tonne)

PRIMARY: BHP ~18/t; Fortescue 19.04/wmt (H1 FY2026); Vale 21.3/t (FY20256 – K); Rio 2026 guidance 23.5–25/wmt. * = Author proxy.



Source: Vale FY2025 6-K (\$21.3/t); BHP FY2025 Ops Review; Fortescue H1 FY2026 (\$19.04/wmt); Rio 2026 guidance (\$23.5–25/wmt).

\$100/dmt CFR · ~5-8% at risk · ~80-130 Mt

China domestic marginal. All major seaborne exporters viable.

\$90/dmt CFR · ~15-20% at risk · ~240-320 Mt

Roy Hill, Kumba at margin. ArcelorMittal Liberia at margin during ramp.

\$80/dmt CFR · ~30-35% at risk · ~480-560 Mt

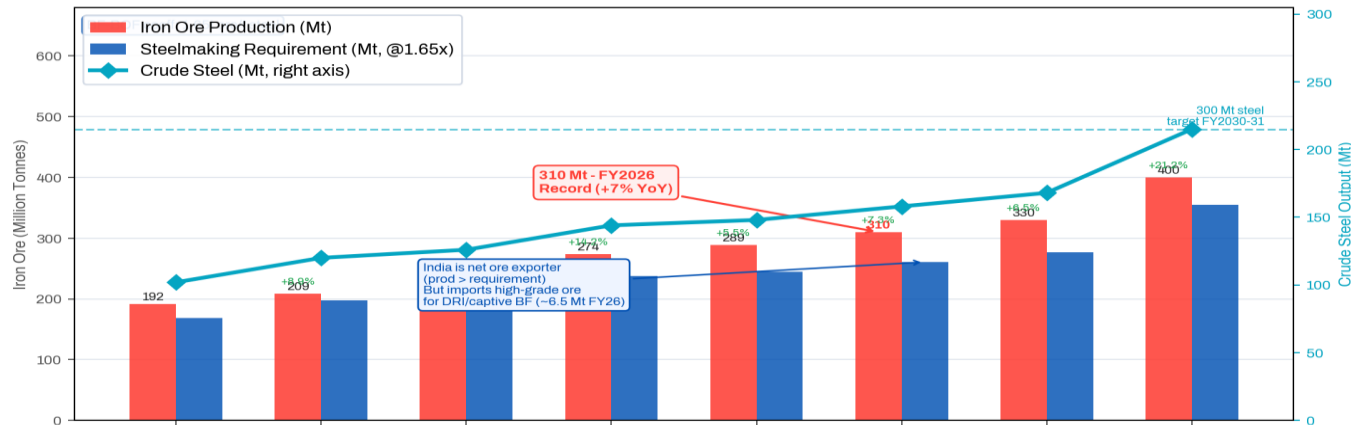
Kumba/Roy Hill loss-making. NMDC exports uncompetitive. China domestic curtailed.

India: FY2026 = 310 mnt output+ Mine Auction Linkage to Global Surplus

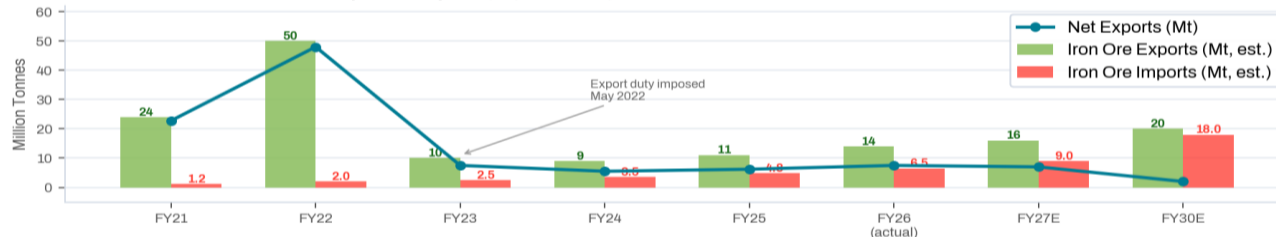
India: Iron Ore Production, Steelmaking Requirement & Trade Balance (FY Basis)

FY2026 = 310 Mt (confirmed record, +7% YoY). Coefficient: 1.65t ore/t steel (BF-BOF route). India is net ore exporter but imports growing volumes of high-grade material for DRI/captive BF.

Panel A - Production vs. Steelmaking Requirement & Crude Steel Output



Panel B - India Iron Ore Trade: Exports & Imports (Mt, estimates)



Source: IBM Ministry of Mines; JPC India; BigMint market data. Trade: DGCIS / Ministry of Commerce estimates. FY27E-FY30E = BigMint Research projections.

310 mnt

India FY2026 actual (slower than steel production growth rate of 11% y-o-y, but still +7% YoY) Apr 14 2026

53 mnt

NMDC FY2026 actual (+21% YoY, record)
OMC: 40 mnt (+11%)

289 mnt

India FY2025 (May 2025)
FY2025 ≠ FY2026 - different years

300 mnt

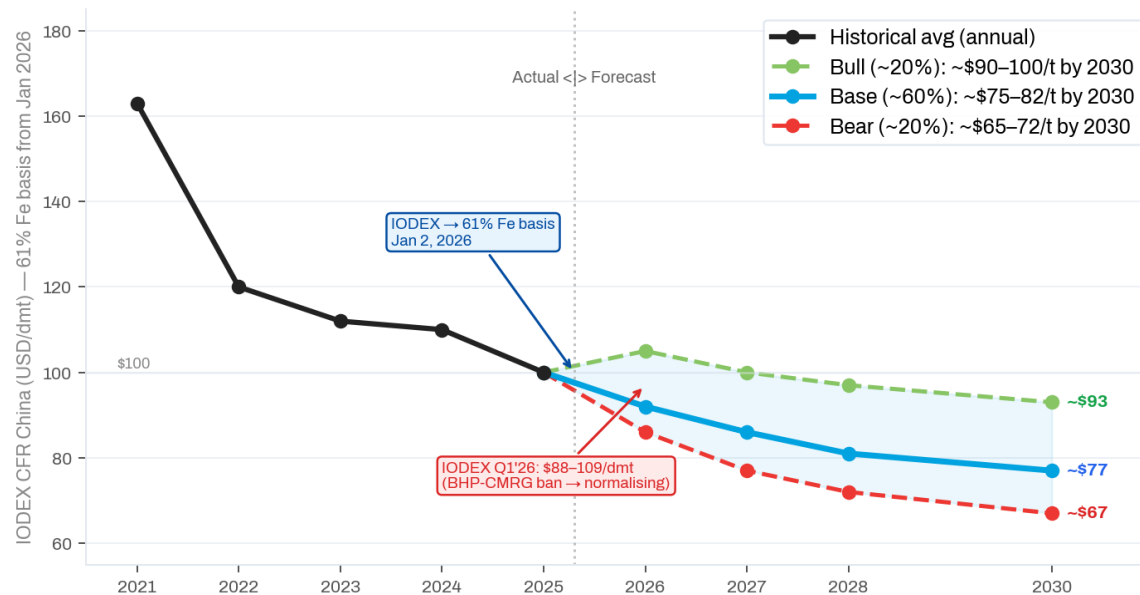
Steel capacity target FY2030-31
Requires ~450-500 Mt ore supply

Mine Auction Linkage: Global surplus → Iron Ore fines index \$75-82/dmt base case → import parity pressure → compression of IBM ASP auction mine economics → IRR sensitivity shifts at auctioned mines → future bid aggressiveness re-rates. Direct link to BigMint Mine Auction Economics study.

IODEX CFR China (61% Fe): Bear / Base / Bull Price Scenarios to 2030

IODEX CFR China Price Outlook: Bear / Base / Bull to 2030 (USD/dmt)

Benchmark: 61% Fe basis from Jan 2, 2026. Supply-conditional; demand held constant.



Source: IODEX market data; BigMint annual average price data 2021–2025; BigMint Research forecasts, April 2026.

61% Fe basis note: Q1 2026 IODEX range: \$88-109/dmt (Jan low to Mar BHP-CMRG Newman ban spike). Transitional 61/62% spread ~\$1.5-2/t through Dec 2027. Scenario ranges supply-conditional.

BULL ~20%

~\$90-100/dmt by 2030

Simandou delayed · China stimulus · India absorbs · CMRG ban sustained

BASE ~60%

~\$75-82/dmt by 2030

Simandou on schedule · China flat to -1% · India +8-12% · Inventory normalises 2027

BEAR ~20%

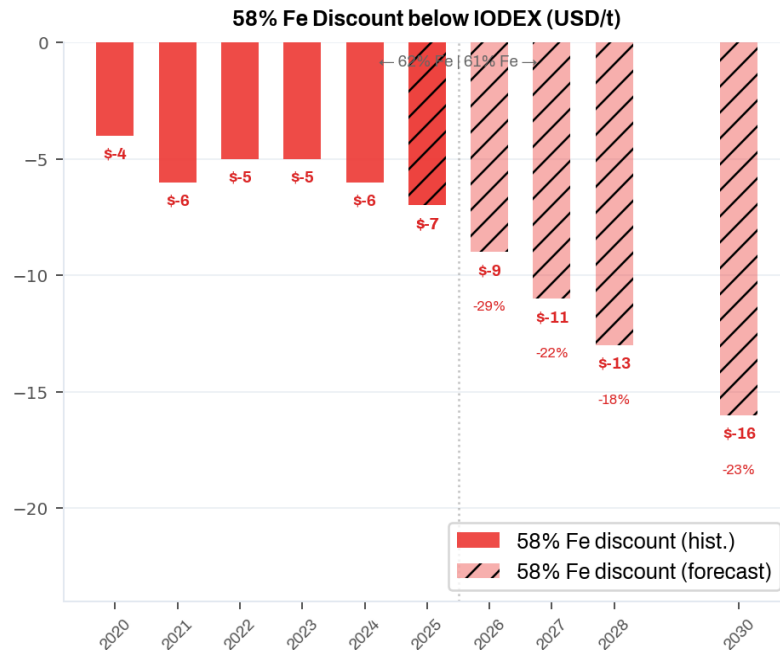
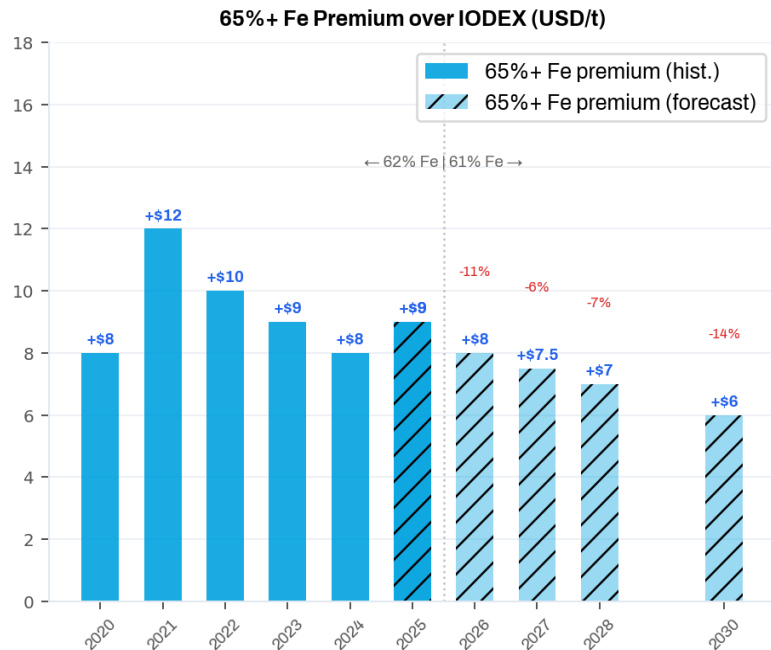
~\$65-72/dmt by 2030

Both Simandou blocks front-load · China -3%+ · Scrap accelerates · Port stocks persist

Grade Bifurcation: DR-Grade Resilient; 58% Fe Discount Widens Structurally

Grade Premium / Discount vs. IODEX Benchmark (USD/t)

Pre-2026: vs. 62% Fe IODEX, 2026+: vs. 61% Fe IODEX (benchmark shift Jan 2, 2026). % labels show YoY spread change.



Source: BigMint price assessments; industry grade differential data. Forecast: BigMint Research, April 2026.

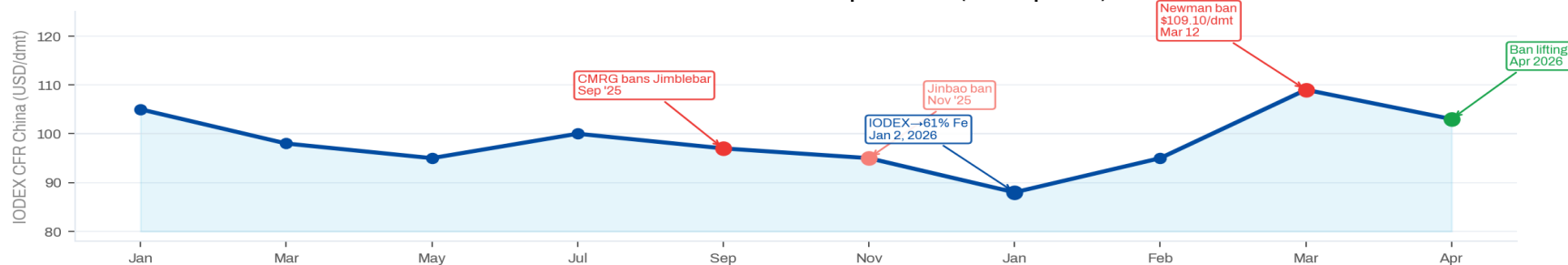
65%+ Fe: Simandou 65.3% Fe is 4.3% above the new 61% Fe benchmark. DRI/EAF buildout supports demand. Premium compresses from ~\$9/t to ~\$5-7/t by 2030 as Simandou scales.

58% Fe: Discount widens from ~\$7/t to ~\$13-16/t by 2030. Mills prefer grade under carbon constraints. Indian BF operators and China domestic ore are most exposed.

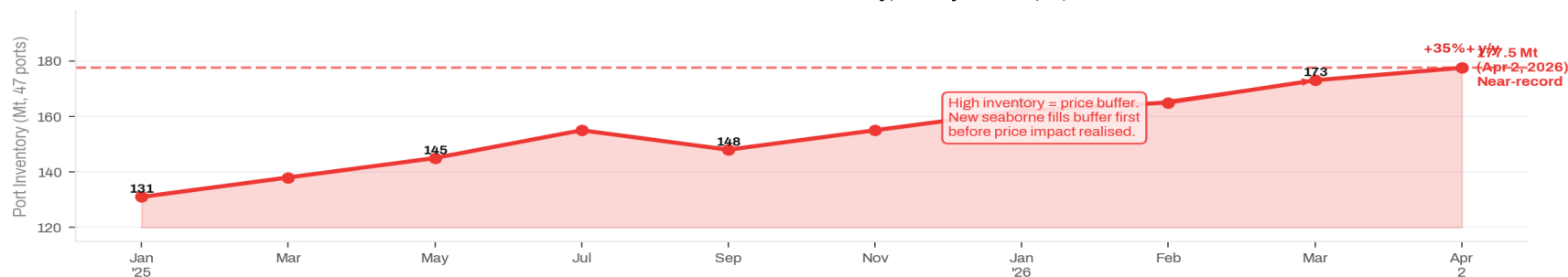
BHP-CMRG Dispute + China Port Inventory: Live Market Context (Apr 2026)

Two developments absent from most 2026 supply outlooks: buyer procurement power and near-record port inventory buffer.

Panel A — IODEX Price & BHP-CMRG Dispute Events (2025–Apr 2026)



Panel B — China Port Iron Ore Inventory, 47 Major Ports (Mt)



Source: IODEX: BigMint market data. Inventory: Mysteel 47-port survey (Apr 2, 2026 = 177.5 Mt). BHP-CMRG: market reports, March 2026; analyst shipment data.

Key insight: 177.5 Mt port inventory (Apr 2 2026, Mysteel) = ~45 days import cover = primary price cap. New seaborne arrivals fill this buffer before price discovery resets. BHP-CMRG ban (lifting Apr'26) adds \$5-10/t near-term upside. Both effects absent from most 2026 supply outlooks.

1. Index is now 61% Fe (from Jan 2, 2026)

All 2026+ price references on 61% Fe basis. Transitional 61/62% spread ~\$1.5-2/t through Dec 2027. Scenario ranges unchanged - the basis difference is not material at a 5-year horizon.

2. The surplus is structural around ~67 mnt net new seaborne in CY2026 alone

Multiple independent supply sources arrive concurrently into plateauing demand. China steel output fell 4.4% in 2025. 177.5 Mt port inventory means Stage 1 is a buffer phase; Stage 2 is the price reckoning.

3. Simandou CY2026 = 25-35 Mt combined (NOT 5-10 Mt)

Rio's 5-10 Mt = SimFer equity only. WCS (Baowu 51%) ramping faster, adds ~22 Mt. Combined working estimate 25-35 Mt (mid 29 Mt). Using Rio's figure understates 2026 impact 3-4x.

4. India FY2026 = 310 mnt output

FY2025 = 289 Mt; FY2026 = 310 Mt (+7% YoY). NMDC = 53 Mt (+21%). Different fiscal years; do not conflate.

5. Base case IODEX \$75-82/dmt by 2030; grade bifurcation the key strategic variable

DR-grade (65%+ Fe) compresses from top but supported by DRI/EAF demand. 58% Fe discount widens to \$13-16/t. India mine auction IRR sensitivity directly linked to this seaborne price trajectory.